
CE 570: Construction Network Scheduling and Cost Management

Course Code: CE 570

Course Name: Construction Network Scheduling and Cost Management

L-T-P-C: 2-0-0-2

Prerequisites: N/A

Intended for: UG

Approval: 64th BoA

Preamble

This course adopts an applied and practical approach to Construction Network Scheduling and Cost Management. The course modules are designed to illustrate several aspects of planning, scheduling, monitoring, and controlling construction projects under time and budget constraints. Students first observe how typical construction projects are organized and represented through network scheduling techniques, then analyze project data to identify critical paths, activity durations, resource allocations, and cost implications. Emphasis is placed on connecting practical project scenarios with idealized analytical models, understanding scheduling logic, time-cost optimization, resource leveling, and project uncertainties directly through case studies and software-based exercises. Students learn to operate scheduling tools, cost estimation software, and project monitoring systems. This hands-on framework develops strong practical understanding, analytical thinking, and managerial insight for applications in building construction, bridges, highways, infrastructure systems, and industrial projects. The course is designed to aid an understanding of construction planning and management problems as relevant to modern engineering applications.

Course Description

The course focuses on:

- Network-based planning (CPM/PERT)
- Scheduling under constraints
- Optimization approaches in construction
- Cost control, earned value analysis, and productivity

Learning Objectives

Students will be able to:

- Develop network schedules
- Analyze critical paths and delays
- Integrate cost and schedule
- Apply optimization concepts
- Interpret project performance

Module-wise Breakdown

- **Module 1: Fundamentals of Project Planning:** Work breakdown structures; Activity definition (2 hrs)
- **Module 2: Network Scheduling Techniques:** CPM / PERT; Critical path analysis (8 hrs)
- **Module 3: Time–Cost Tradeoffs:** Crashing; Cost optimization (6 hrs)
- **Module 4: Resource Allocation & Optimization:** Resource leveling; Linear Programming Applications (6 hrs)
- **Module 5: Productivity & Performance Control:** Monitoring; Earned value analysis (6 hrs)

Delivery Format

- Lectures with worked examples and case-based discussions
- Short assignments / exercises

Total: 28 hours